

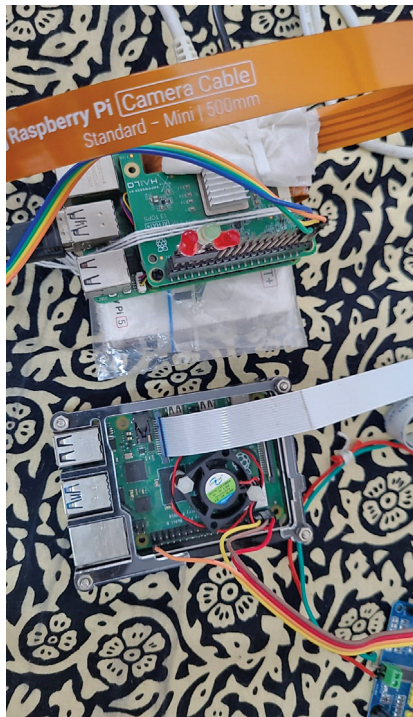


Fig. 4: Experimental setup of Raspberry Pi 5 with camera and Hailo-8L AI accelerator

pan-tilt angles, eliminating the need for an HDMI display.

Software coding and installation

All software used in the system is open source. The primary operating system is Debian GNU/Linux 12 (Bookworm). A 64-bit operating system is required and is available at <https://www.raspberrypi.com/documentation/computers/ai.html#getting-started>

For installing Hailo-8L, refer to [https://github.com/hailo-ai/hailo-rpi5-examples/blob/main/ README.md#installation](https://github.com/hailo-ai/hailo-rpi5-examples/blob/main/README.md#installation)

First, download the repository to the Raspberry Pi 5:

```
$ git clone https://github.com/hailo-ai/hailo-rpi5-examples.git
$ cd hailo-rpi5-examples
```

This directory contains two important files: `setup_env.sh` and `requirements.txt`. Hailo-8L requires Python packages of specific versions. Mixing these packages with a common Python installation may result in conflicts. To avoid this, a dedicated virtual environment is recommended, allowing the required

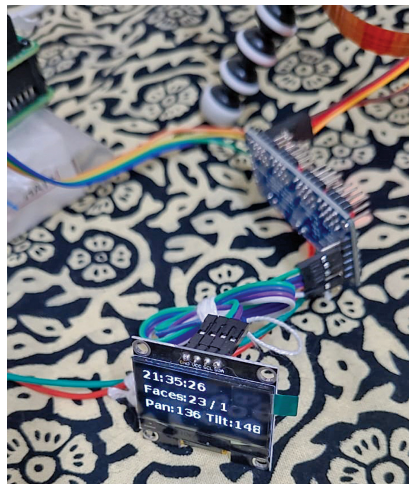


Fig. 5: PC OLED displaying face count and pan-tilt angles

package versions to be installed without affecting the system-wide Python installation.

Activate the virtual environment:

```
$ source setup_env.sh
```

Once activated, the command prompt changes to indicate the active virtual environment:

```
(venv_hailo_rpi5_examples)
bera@raspberrypi:~/hailo-rpi5-examples $
```

Inside the virtual environment, install the required packages:

```
(venv_hailo_rpi5_examples)
bera@raspberrypi:~/hailo-rpi5-examples
$ pip install -r requirements.txt
```

The `requirements.txt` file contains several package entries that can also be installed individually.

```
$ cat requirements.txt
```

Typical entries include:

```
numpy<2.0.0
setproctitle
opencv-python
```

If required, each package can be installed separately:

```
$ pip install "numpy<2.0.0"
$ pip install setproctitle
$ pip install opencv-python
```

To download the required Hailo models, run:

```
$ ./download_resources.sh
```

For installing additional Python code: <https://github.com/hailo-ai/hailo-apps-infra>

This repository provides additional Hailo Applications Infrastructure for the Raspberry Pi 5, parts of

which are optional for this system.

Clone the repository:

```
$ git clone https://github.com/hailo-ai/hailo-apps-infra.git
```

Navigate to the directory and run the optional installation:

```
$ cd hailo-apps-infra
$ pip install --force-reinstall -v -e .
# [optional]
```

This step installs additional Python code for experimentation and can be skipped if not required. At this stage, all necessary system software is installed and ready for use.

Face-tracking code is developed in the following directory:

```
$ cd /home/bera/hailo-rpi5-examples/hailo-apps-infra/hailo_apps_infra/
```

Common items for Raspberry Pi 5 and Raspberry Pi 4

Since the system uses an I²C OLED display and a PCA9685 servo driver board, Python must be configured to recognise these devices correctly. Run the following commands inside the virtual environment:

```
$ pip install luma.oled
$ pip install adafruit-circuitpython-servokit
```

Hailo-8L python code and operation

The following additional Python files are used in the system and become accessible once the virtual environment is active:

- **`hailo_rpi_common.py`** –

Contains utility functions and shared methods and is loaded automatically from the virtual environment path

- **`fdetection_pipeline.py`** –

Defines the face-detection pipeline and must reside in the working directory

- **`personface_labels.json`** –

Provides labels for detected faces and must remain in the same directory as the main program

- **`pi5_facetracking.py`** –

The main Python file that launches the real-time face-tracking application

To display command-line help: